CHANGHEON HAN

Institute for Physical Artificial Intelligence (IPAI) Postdoctoral Fellow
School of Aeronautics and Astronautics, Purdue University, West Lafayette, IN, USA
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RESEARCH INTERESTS

- Generalizable & Human-Centered Physical AI for Manufacturing Automation
- Human-Robot Collaboration & Multi-Robot Systems for In-Space Servicing, Assembly, and Manufacturing (ISAM)
- Autonomous In-Situ Resource Utilization (ISRU) Systems for Planetary ISAM

EDUCATION

Purdue University — **West Lafayette, Indiana, USA**
Ph.D. in Mechanical Engineering — 2022-2026
- Dissertation: Generalizable and Edge-Deployable Industrial AI via Foundation Model, Lightweight Model, and Generative Multimodal Monitoring
- Advisor: Martin B.G. Jun

Sogang University — **Seoul, South Korea**
M.S. in Mechanical Engineering — 2013-2015
- Thesis: Hydrophobicity Effects on Water Transport through Aquaporin-mimic Nanoporous Membranes
- Advisor: Daejoong Kim

Sogang University — **Seoul, South Korea**
B.S. in Mechanical Engineering (Magna Cum Laude) — 2009-2013
- Topic: Energy Harvesting System Using Reverse Electrodialysis with Nanoporous Membranes
- Advisor: Daejoong Kim

AFFILIATIONS & APPOINTMENTS

Purdue University — **West Lafayette, IN, USA**
Institute for Physical Artificial Intelligence (IPAI) Postdoctoral Fellow — 2026-Present
- Physical AI for human-autonomy teaming and robotic ISAM (Mentor: Shaoshuai Mou)

Poseidon — **Palo Alto, CA, USA**
AI Research Fellow — 2025-2026
- Deepfake detection, Automatic Speech Recognition (ASR) and Text-to-Speech (TTS) models

Purdue University — **West Lafayette, IN, USA**
Ward A. Lambert Teaching Fellow — 2025
- Instructor of Record: ME 20000 Thermodynamics I

Korea Aerospace Research Institute (KARI) — **Daejeon, South Korea**
Researcher — 2015-2021
- KSLV-II (Korean Space Launch Vehicle-II, Nuri) Propulsion System Development and Launch Program Operation
- Space Environmental Tests for the KSLV-II, the KPLO (Korea Pathfinder Lunar Orbiter, Danuri), and Satellites
- Mass Property Tests for the KF-21 Boramae
- Mixed Reality Technology for Spacecraft Design Applications (PI, Collaborator: Microsoft)

SELECTED AWARDS & HONORS

NSF Manufacturing Blue Sky Competition Finalist, *SME NAMRC54*	2026
Institute for Physical Artificial Intelligence Postdoctoral Fellowship, *Purdue University*	2026–2028
Best Poster Award (1st Place), *US-Korea Conference (UKC)*	2025
KSEA-KUSCO Graduate Scholarship, *Korean-American Scientists and Engineers Association (KSEA)*	2025
Travel Grant Award, *Purdue University*	2025
Estus H. and Vashti L. Magoon Graduate Student Research Excellence Award, *Purdue University*	2025
Bilsland Dissertation Fellowship, *Purdue University*	2025–2026
Best Poster Award, *DigiTwin 2024 for Manufacturing Sustainability, Safety, and Resilience*	2024
NSF Manufacturing Blue Sky Competition Finalist, *SME NAMRC52*	2024
Travel Grant Award, *Purdue University*	2024
Ward A. Lambert Graduate Teaching Fellowship, *Purdue University*	2024–2025
Top 1% Students with Excellent Grades, *Sogang University*	2012
Academic Excellence Scholarship and Honors, *Sogang University*	2009–2010 & 2012

SELECTED PEER-REVIEWED JOURNAL PUBLICATIONS

1. **Changheon Han**, Yun Seok Kang, Yuseop Sim, Hyung Wook Park, Martin B.G. Jun, "LISTEN: Lightweight Industrial Sound-representable Transformer for Edge Notification", *Advanced Engineering Informatics*, 2026.
2. Hojun Lee, **Changheon Han**, Ted Gabor, Yuseop Sim, Martin Byung-Guk Jun, Yongho Jeon, Semih Akin, Jiho Lee, "Cold Spray-Based Secure and Unique Product Identification with Neural Encoding: A Full-Stack Framework for Scalable Authentication in Manufacturing", *Journal of Intelligent Manufacturing*, 2026.
3. Jurim Jeon, Yuseop Sim, Hojun Lee, **Changheon Han**, Dongjun Yun, Eunseob Kim, Shreya Laxmi Nagendra, Yangjin Kim, Sang Won Lee, Martin B.G. Jun, Jiho Lee, "ChatCNC: Conversational Machine Monitoring via Large Language Model and Real-Time Data Retrieval Augmented Generation", *Journal of Manufacturing Systems*, 2025.
4. **Changheon Han**, Jiho Lee, Martin B.G. Jun, Sang Won Lee, Huitaek Yun, "Visual coating inspection framework via self-labeling and multi-stage deep learning strategies", *Journal of Intelligent Manufacturing*, 2025.
5. **Changheon Han**, Jiho Lee, Hojun Lee, Yuseop Sim, Jurim Jeon, Martin B.G. Jun, "Zero-shot Autonomous Robot Manipulation via Natural Language", *Manufacturing Letters*, 2024.
6. Fengfeng Zhou, Xingyu Fu, Siying Chen, **Changheon Han**, Martin B.G. Jun, "3D Profile Reconstruction and Internal Defect Detection of Silicon Wafers Using Cascaded Fiber Optic Fabry-Pérot Interferometer and Leaky Field Detection Technologies", *Journal of Manufacturing Science and Engineering*, 2024.
7. **Changheon Han**, Heebum Chun, Huitaek Yun, Jiho Lee, Fengfeng Zhou, ChaBum Lee, Martin B.G. Jun, "Hybrid Semiconductor Wafer Inspection Framework via Autonomous Data Annotation", *Journal of Manufacturing Science and Engineering*, 2024.
8. Jinho Park, **Changheon Han**, Martin B.G. Jun, Huitaek Yun, "Autonomous robotic bin picking platform learning from human demonstration and YOLOv5", *Journal of Manufacturing Science and Engineering*, 2023.
9. **Changheon Han**, Dai Tang, Daejoong Kim*, "Molecular Dynamics Simulation on the Effect of Pore Hydrophobicity on Water Transport through Aquaporin-mimic Nanopores", *Colloids and Surfaces A*, 2015.
10. Kilsung Kwon, Seung Jun Lee, Longnan Li, **Changheon Han**, Daejoong Kim, "Energy Harvesting System using Reverse Electrodialysis with Nanoporous Polycarbonate Track-etch Membranes", *International Journal of Energy Research*, 2014.

SELECTED PEER-REVIEWED CONFERENCE PAPERS

1. Hojun Lee, Yuseop Sim, **Changheon Han**, Jiho Lee, Aniket Bera, Changju Kim, Martin B.G. Jun, "EASEIR: Efficient and Adaptive Safe-set Estimation via Implicit Representation for High-dimensional Motion Planning", *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2025.
2. **Changheon Han**, Ted Gabor, Hojun Lee, Jiho Lee, Semih Akin, Martin B.G. Jun, "Pulsed Cold Spray System for Physical Unclonable Function Generation", *Procedia CIRP*, 2025.
3. **Changheon Han**, Hojun Lee, Jiho Lee, Martin B.G. Jun, Huitaek Yun, "A Study on Digital Twin for Autonomous Robotic Chip Removal Deep Learning Framework on CNC Machine", *International Manufacturing Science and Engineering Conference (MSEC)*, 2024.
4. **Changheon Han**, Heebum Chun, ChaBum Lee, Martin B.G. Jun, "Stroboscopic Data-Driven, Integrated, and Intelligent Machine Learning-Based Algorithms for Semiconductor Wafer Inspection", *International Manufacturing Science and Engineering Conference (MSEC)*, 2023.

PREPRINT & PAPERS UNDER REVIEW

1. **Changheon Han**, Martin B.G. Jun, "Narrative Manufacturing: From Mass Production to Mass Innovation", *Manufacturing Letters*, 2026. (Under review)
2. Junsik Nam, Hojun Lee, **Changheon Han**, Yuseop Sim, Ragu Athinarayanan, Martin B.G. Jun, Jiho Lee, "C3AM: Conversational CAD/CAM Automation via Multi-modal Large Language Model Agents", *Journal of Intelligent Manufacturing*, 2026. (Under revision)
3. **Changheon Han**, Yuseop Sim, Hoin Jung, Jiho Lee, Hojun Lee, Yun Seok Kang, Sucheol Woo, Garam Kim, Hyung Wook Park, Martin Byung-Guk Jun, "IMPACT: Industrial Machine Perception via Acoustic Cognitive Transformer", *arXiv:2507.06481*, 2025.

INTELLECTUAL PROPERTIES

1. Martin B.G. Jun, **Changheon Han**, Yuseop Sim, "End-to-End On-device Real-time Acoustic Condition Monitoring System leveraging Lightweight Industrial Foundation Model," US Patent (In preparation), 2025
2. Martin B.G. Jun, Jiho Lee, Hojun Lee, **Changheon Han**, Yuseop Sim, Semih Akin, Ted Gabor, "Cold Spray-Enabled Physically Unclonable Identifier and its Spectral Authentication via Implicit Neural Representation," US Patent (Filed), 2025
3. **Changheon Han**, Junseong Lee, "Method of Analyzing Fluid Flow," Korea Patent, Registration No.1025471900000, Jun 2023
4. Chang-lae Cho, Jong-Min Im, Hee-Kwang Eun, Jong-Hyub Jun, Namjin Moon, **Changheon Han**, "Clamp band tensioning device and tensioning method," Korea Patent, Registration No. 1024687450000, Jun 2022

TEACHING EXPERIENCE

Purdue University — **West Lafayette, IN, USA**

- Instructor of Record (Ward A. Lambert Teaching Fellow), *ME 20000: Thermodynamics I*, Spring 2025
- Lab Instructor, *ME 26300: Introduction to Mechanical Engineering Design*, Spring 2023

Sogang University — **Seoul, South Korea**

- Graduate Teaching Assistant, *MEE 3025: Mechanical Engineering Experiments I*, Fall 2013
- Graduate Teaching Assistant, *MEE 6452: Internal Combustion Engine*, Fall 2013
- Graduate Teaching Assistant, *MEE 4021: Creative Capstone Design*, Spring 2013
- Graduate Teaching Assistant, *MEE 2022: Thermodynamics I*, Spring 2013

FUNDING

1. Institute for Physical Artificial Intelligence Postdoctoral Fellowship, *Purdue University*, 2026-2028 ($175,000)
2. Bilsland Dissertation Fellowship, *Purdue University*, 2025-2026 ($73,000)
3. Ward A. Lambert Graduate Teaching Fellowship, *Purdue University*, 2024-2025 ($75,400)
4. (PI) Developing NewSpace Design Process of Aerospace by Using Physics Engines and Mixed Reality, *Ministry of Science and ICT*, 2020 ($50,000)
5. (PI) Developing Efficient Design Process of Launch Vehicle by Using Physics Engines and Mixed Reality, *Ministry of Science and ICT*, 2019 ($33,000)

PROJECT PARTICIPATION

1. Human Centric, Resilient, and Sustainable Cyber Physical System (CPS) for Future Factories, *Hitachi American Ltd.*, 2024-2026
2. Development of AI-Based Robot Technology for Machining Chip Recognition and Removal, *Korea Institute of Machinery & Materials (KIMM)*, 2024-2026
3. Immersive & Interactive Cyber-Physical System Utilizing AI/ML and VR to Train, Operate and Optimize Machine Vision Guided Robots, *U.S. Air Force Research Laboratory*, 2023-2026
4. Collaborative Research: Data-Driven Metrology and Inspection Technology for Semiconductor Wafer-Level Manufacturing, *U.S. National Science Foundation (NSF)*, 2022-2023
5. Proof of Concept Assessment for Automated Visual Inspection, *Purdue Technical Assistance Program*, 2022
6. Development of the GEO-KOMPSAT-3, *Ministry of Science and ICT (MSIT)*, 2021
7. Development of Pathfinder Lunar Orbiter and Key Technologies for the Second Stage Lunar Exploration, *MSIT*, 2021
8. System and Bus Development of the Seventh Korea Multi-Purpose Satellite, *MSIT*, 2020-2021
9. Korean Fighter eXperimental (KF-X), *Republic of Korea Air Force*, 2020-2021
10. System Integration and Development of the Sixth Korea Multi-Purpose Satellite, *MSIT*, 2020
11. Developing NewSpace Design Process of Aerospace by Using Physics Engines and Mixed Reality, *MSIT*, 2020
12. Developing Efficient Design Process of Launch Vehicle by Using Physics Engines and Mixed Reality, *MSIT*, 2019
13. Korea Space Launch Vehicle II (KSLV-II) Program, *MSIT*, 2015-2019
14. Analysis of Water Transport Property and System Design, *MSIT*, 2013-2014
15. Multi-Phenomena CFD Engineering Research Center, *MSIT*, 2013-2014
16. Study of Bottom-Up Methodology for Multiscale CFD, *MSIT*, 2012
17. Osmotic Seawater Desalination Using Nanochannels, *MSIT*, 2011-2013

SERVICE & LEADERSHIP

Reviewer, *International Journal of Production Research*
Reviewer, *Scientific Reports*
Seminar Organizing Committee, *Purdue Institute for Control, Optimization and Networks (ICON)*, 2026-Present
Vice President, *Purdue Graduate Association for Manufacturing Excellence*, 2025
Volunteer, *NAMRC and MSEC*, 2022
Founder & President, *Korea Aerospace Research Institute Junior Board*, 2021

INVITED TALKS & SEMINARS

"Generalizable and Deployable Industrial AI for Real-World Manufacturing", *Korea University*, 2026
"Narrative Manufacturing", *NSF Manufacturing Blue Sky Competition*, 2026
"Manufacturing Innovation in Aerospace", *KARI*, 2025
"Introduction to Smart Manufacturing Technologies", *MMRL Student Symposium*, 2025
"Human-Robot Collaborative Smart Manufacturing Leveraging Digital Twin", *FLEX Lab Seminar*, 2025
Robotics Showcase and Demo Day, *Senate AI Caucus*, 2024

PROFESSIONAL AFFILIATIONS

Society of Manufacturing Engineers (SME), Member (joined 2025)
American Society of Mechanical Engineers (ASME), Member, Member (joined 2025)
Korean-American Scientists and Engineers Association (KSEA), Member, Member (joined 2024)
Sigma Xi, Member (Joined 2026)

MENTORSHIP EXPERIENCE

Graduate Students

- Wenxi Chen, *Ph.D. Student, Aeronautics and Astronautics, Purdue University*, 2026–Present
- Monish Lokhande, *Ph.D. Student, Aeronautics and Astronautics, Purdue University*, 2026–Present
- Sidhdharth Sikka *Ph.D. Student, Aeronautics and Astronautics, Purdue University*, 2026–Present
- Hoin Jung, *Ph.D. Candidate, Electrical and Computer Engineering, Purdue University*, 2025–2026
- Dong Jae Lee, *Ph.D. Student, Earth, Atmospheric, and Planetary Sciences, Purdue University*, 2025–2026
- Kaeul Lim, *Ph.D. Candidate, Mechanical Engineering, Purdue University*, 2025–2026 *(currently Postdoctoral Research Associate at Purdue University)*
- Hayoung Jeong, *M.S. Student, Mechanical Engineering, Purdue University*, 2024–2025 *(currently at Ultium Cells)*
- Hojun Lee, *Ph.D. Student, Mechanical Engineering, Purdue University*, 2023–2025
- Yuseop Sim, *Ph.D. Student, Mechanical Engineering, Purdue University*, 2023–2025

Undergraduate Students

- Taejoon Hong, *Aeronautics and Astronautics, Purdue University*, 2025–2026
- Seong Bin Choi, *Mechanical Engineering, Purdue University*, 2025 *(currently Ph.D. Student at Purdue University)*
- Jong Yun Won, *Computer Science, Purdue University*, 2025 *(currently at Korea Institute of Industrial Technology)*
- Jaewook Lee, *Electrical and Computer Engineering, Purdue University*, 2025

Professional and Pre-College Mentoring

- Rashimi Nagpal, *AI Engineer, Poseidon*, 2025–2026 *(currently Ph.D. Student at MIT CSAIL)*
- Ellen Kim, *High School Student*, 2023–2024 *(currently Undergraduate Student at University of Southern California)*